MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS

MATH 1101- PRECALCULUS

SUPPLEMENTARY EXAMINATION

Date: 16th March, 2018 Time Allowed: 3 hours

INSTRUCTIONS

- Check that the paper has **13** questions
- Attempt **all** questions
- Show all your working

1. Evaluate
   (a) $\sqrt[4]{a^3b}\sqrt[4]{ab^3}$ [3]
   (b) $\sqrt[3]{-8}$ [2]
2. Rationalise the denominator.
   (a) $\dfrac{3}{\sqrt[4]{3}}$ [3]
   (b) $\dfrac{\sqrt{2}}{1-\sqrt{2}}$ [4]
3. Solve the equation $|x-5|=1$. [4]
4. Let $g(x)=2x+1$. Find $f(x)$ given that $g(f(x))=10x-7$. [4]
5. If $f(x)=3x-1$, find $f^{-1}(x)$. [5]
6. Sketch the graph of $y=x^2+3$ without using the table of values. [4]
7. Solve the inequality $x^2-3x-4\geq 0$. [7]
8. Determine the domain and range of the function $y=4x-5$. [4]
9. Let $\log_b 2=A$, $\log_b 3=B$ and $\log_b 5=C$. Evaluate
   (a) $\log_b 15$. [2]
   (b) $\log_b 0.3$. [3]
10. Solve the equation $\log_2(x^2-4)=5$ [4]
11. (a) Convert $150°$ to radians. [2]
    (b) Convert $\dfrac{2\pi}{3}$ to degrees. [2]
12. Evaluate each expression
    (a) $\sin 600°$. [4]
    (b) $\sec\left(-\frac{\pi}{4}\right)$. [4]
13. Prove $\dfrac{\sin\theta}{1+\cos\theta}+\dfrac{1+\cos\theta}{\sin\theta}=2\csc\theta$ [5]